QF600 Asset Pricing — Week 5 Reference Cheatsheet

Advanced Portfolio Construction: estimation risk, factor-neutral portfolios, index tracking

Built from Lecture 5 slides (Liu Peng, LKCSB) · September 2026

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*Source: "Lecture slides – Week 5 v4" (Liu Peng, LKCSB, 48 slides). Slide numbers are PDF page numbers and are shown as [S#]. Material not on the slides is tagged **[Beyond slides]**. ⚠ marks slide ambiguities or errors, with the corrected version.*

*Exam context (Week 1 [S5]): the final is closed book, with one double-sided A4 sheet allowed. It is 40% MCQ and 60% written (concepts plus simple calculations), with **no coding**. The Python here is for the group homework and for checking your own understanding. What you need in the exam is the formulas, the derivations and 2-asset hand calculations.*

Notation used throughout (slide conventions, [S4])

Symbol	Meaning	Dim / units	Notes
N	number of assets	scalar	
K	number of factors (Part 1–2); **also** max number of names held (Part 3, $\|w\|_0 \le K$)	scalar	⚠ the slides use K for both. Context decides which.
w	portfolio weights (fraction of NAV in each asset)	$N \times 1$	Sums to 1 if fully invested; negative = short
w^-	current (drifted, pre-trade) holdings	$N \times 1$	
μ ($\hat{\mu}$)	expected returns (estimate)	$N \times 1$, return/period (annual, decimal)	⚠ In [S5], μ also denotes $E[\tilde{f}]$ (factor means). Don't mix the two.
$\tilde{\mu}$	a candidate "true" mean in the uncertainty set U	$N \times 1$	
Σ ($\hat{\Sigma}$)	covariance of asset returns (estimate)	$N \times N$, return^2	
λ	risk-aversion penalty	scalar > 0, units 1/return	⚠ In [S29], $\mu = B\lambda + \alpha$ uses λ as the $K \times 1$ **vector of factor risk premia** $= E[\tilde{f}]$. That is a different object. Below I write it λ_f when the two could be confused.
η	ridge (L2) penalty strength	scalar ≥ 0, return units	
ε / κ	radius of the mean-uncertainty ellipsoid	scalar ≥ 0, dimensionless	[S20] writes ε and the code [S21] writes κ. They are the same quantity.
S	shape matrix of the uncertainty ellipsoid	$N \times N$	The code sets $S = \Sigma$
t	$\sqrt{\hat{\mu}^\top S^{-1} \hat{\mu}}$, "size of the forecast"	scalar	With $S = \Sigma$ it is the **maximum Sharpe ratio** [Beyond slides]
τ	proportional trading-cost rate	return per unit traded, e.g. 20 bp = 0.0020	

Symbol	Meaning	Dim / units	Notes
$\tilde{r}$	asset returns (random)	$N \times 1$	Tilde = random variable
$\tilde{f}$	factor returns	$K \times 1$	
B	factor loadings (betas)	$N \times K$	Row i = asset i's betas
Ω	factor covariance $\mathrm{Cov}(\tilde{f})$	$K \times K$	
$\tilde{\varepsilon}$	idiosyncratic shocks	$N \times 1$	Not to be confused with the radius ε
D	residual covariance $\mathrm{Cov}(\tilde{\varepsilon})$	$N \times N$, diagonal	
α	residual mean (the part of the forecast the factors don't explain)	$N \times 1$	
ν	Lagrange multiplier on $B^\top w = 0$	$K \times 1$	
b	benchmark weights	$N \times 1$	
$a = w - b$	active weights	$N \times 1$	$\mathbf{1}^\top a = 0$ when both are fully invested
$e = a^\top \tilde{r}$	active return	scalar	
TE	tracking error $\sqrt{a^\top \Sigma a}$	return/period	
$\|w\|_2$, $\|w\|_1$, $\|w\|_0$	L2 length, sum of absolute values, number of nonzeros	scalar	

Lower case = vector, upper case = matrix, $^\top$ or $'$ = transpose [S4]. **Implicit convention [Beyond slides]:** the unconstrained MVO $w^* = \frac{1}{\lambda}\Sigma^{-1}\mu$ has no budget constraint. The leftover $1 - \mathbf{1}^\top w$ sits in the risk-free asset, so μ should be read as **excess** returns. This matches Week 3–4, where $\tilde{R}_i - R_f = \alpha_i + \beta_i^\top \tilde{F} + \tilde{\varepsilon}_i$.

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0. Topic map

```
PREREQUISITES
 W1 Expected utility → mean–variance utility  μᵀw − (λ/2)wᵀΣw   (λ = risk aversion)
 W2 Efficient frontier, tangency portfolio, two-fund separation, MVO w* = (1/λ)Σ⁻¹μ,
    estimators μ̂= (1/T)Σr_t, σ² = (1/(T−1))Σ(r_t−μ̂)²
 W3 CAPM: μ_i = R_f + β_i(μ_M − R_f); regression form, α (Jensen), β
 W4 K-factor model  R̃_i − R_f = α_i + Σ_k β_ik F̃_k + ε̃_i ; factors as hidden shared bets
        |
        ▼
WEEK 5: ADVANCED PORTFOLIO CONSTRUCTION  [S1–S7]
 ├─ Portfolio in factor form [S5]: r̃_p = α_p + β_pᵀf̃ + ε̃_p,  β_p = Bᵀw,
 │     Var = β_pᵀΩβ_p + wᵀDw   ← used by Parts 2 and 3
 ├─ Mandates [S7]: absolute return | factor neutral | index tracking
 │
 ├─ PART 1  The input problem (estimation risk)  [S8–S24]
 │   ├─ Error maximizer: Δw = (1/λ)Σ⁻¹Δμ  [S8–S10]
 │   ├─ Fix 1  Ridge / L2:   w* = (λΣ̂+ ηI)⁻¹μ̂  [S12–S18]
 │   │     ├─ L2 norm [S13];  prefers spread-out weights [S16]
 │   │     └─ no signal + budget → 1/N  [S17]
 │   ├─ Fix 2  Robust (ellipsoidal U):  w* = w₀(1 − κ/t)⁺   [S19–S21]
 │   └─ Fix 3  Trading costs, L1 turnover penalty → no-trade region  [S22–S24]
 │
 ├─ PART 2  Factor-neutral portfolios  [S25–S37]
 │   ├─ r̃ = α + Bf̃ + ε̃(skill / factor / luck)  [S26]
 │   ├─ Beta-neutral ⇔ Bᵀw = 0  [S27];  cost: gives up factor premium  [S28]
```

```
│    ├─ Residual space: μ↛α, Σ↛D;  Σ = BΩBᵀ + D  [S29–S30]
│    ├─ Problem max αᵀw − (λ/2)wᵀDw s.t. Bᵀw = 0; Lagrangian solution  [S31–S32]
│    ├─ Constrained optimum = projection in the risk metric  [S33–S34]
│    ├─ Worked example and code  [S35–S36]
│    └─ Feasibility: long-only can't be neutral → relative neutrality Bᵀ(w−b)=0  [S37]
│
├─ PART 3  Index tracking  [S38–S45]
│    ├─ Why track, S&P 500 practice  [S39–S40]
│    ├─ Active weights a = w − b, active return e = aᵀr̃, TE = √(aᵀΣa)  [S41–S42]
│    ├─ TE² = (Bᵀa)ᵀΩ(Bᵀa) + aᵀDa  (factor bets + stock-specific)  [S43–S44]
│    │     └─ links back to Part 2: Bᵀ(w − b) = 0 kills factor bets
│    └─ Sparse replication: min TE² + τ‖w − w⁻‖₁, 1ᵀw=1, w≥0, ‖w‖₀≤K (nonconvex)  [S45]
│
└─ Synthesis table [S46] · In-class quiz [S47] · Group homework pipeline [S48]
       │
       ▼
LATER: W6 SDF/state prices (the premium λ_f that neutral books give up is priced there)
```

Dependency logic:

- **Part 1** is about *how much to trust* μ, and all three fixes shrink the trades.
- **Part 2** is about *which part of* μ *you want paid for*: keep α and hedge $B\lambda_f$.
- **Part 3** is about *a different objective*: stay close to b instead of maximising utility.
- Parts 2 and 3 both rest on $\Sigma = B\Omega B^\top + D$ and the fact that only $B^\top w$ (or $B^\top a$) carries factor risk.
- Trading costs ($\tau\|\cdot\|_1$) appear in Part 1 and again in Part 3.

---

1. Concept entries

1.1 Portfolio in factor form [S5]

- **Definition**: stack the one-asset factor model $\tilde{r} = \alpha + B\tilde{f} + \tilde{\varepsilon}$ across holdings. Then

$$\tilde{r}_p = w^\top \tilde{r} = \underbrace{w^\top \alpha}_{\alpha_p} + \underbrace{(B^\top w)^\top}_{\beta_p^\top} \tilde{f} + \underbrace{w^\top \tilde{\varepsilon}}_{\tilde{\varepsilon}_p}, \qquad \beta_p = B^\top w \ (K \times 1).$$

- **Intuition**: a portfolio is itself a "one asset" with its own alpha, betas and residual. Exposures are **linear in weights**, so they simply add.
- **Formula / statement**:

$$E[\tilde{r}_p] = \alpha_p + \beta_p^\top \mu_f, \qquad \mathrm{Var}(\tilde{r}_p) = w^\top \Sigma w = w^\top (B\Omega B^\top + D) w = \beta_p^\top \Omega \beta_p + w^\top D w.$$

  - $\mu_f = E[\tilde{f}]$. The slide writes this as "μ" ⚠; it is the same as λ_f in [S29].
  - $E[\tilde{\varepsilon}_p] = 0$.
  - $\beta_p^\top \Omega \beta_p$ is **systematic factor risk**, set entirely by $B^\top w$.
  - $w^\top D w = \sum_i w_i^2 d_i$ is **residual luck**. It diversifies as N grows.
- **Assumptions**:
  - $\mathrm{Cov}(\tilde{f}, \tilde{\varepsilon}) = 0$, so the cross terms drop.
  - D is diagonal, i.e. residuals are uncorrelated across stocks.
  - If either assumption fails, the variance picks up extra $2w^\top B\,\mathrm{Cov}(\tilde{f}, \tilde{\varepsilon})w$ or off-diagonal D terms, and "residual risk diversifies" is no longer guaranteed.
- **Derivation**: $\mathrm{Var}(w^\top B\tilde{f} + w^\top \tilde{\varepsilon}) = w^\top B\Omega B^\top w + w^\top D w + 2w^\top B\,\mathrm{Cov}(\tilde{f}, \tilde{\varepsilon})w$, and the last term is 0.
- **Worked example [Beyond slides]**:
  - Setup: one factor with $\sigma_f^2 = \Omega = 0.04$, $B = (1.2, 0.8)^\top$, $D = \mathrm{diag}(0.02, 0.01)$, $w = (0.6, 0.4)$.
  - Exposure: $\beta_p = 0.72 + 0.32 = 1.04$.
  - Factor variance: $1.04^2 \times 0.04 = 0.043264$.

  - Residual variance: $0.36(0.02) + 0.16(0.01) = 0.0072 + 0.0016 = 0.0088$.
  - Total: Var = 0.052064, so $\sigma_p = 22.82\%$. Factor share = 83.1%.

- **Interpretation**:
  - Equal weights $1/N$ give residual variance $\frac{1}{N}\bar{d} \to 0$, but factor risk stays at $\bar{\beta}^\top \Omega \bar{\beta}$.
  - Diversification kills luck, not factor risk.

- **Traps**:
  - Don't add *variances* of individual factor bets asset by asset. Aggregate exposures first ($B^\top w$), then apply Ω.
  - $w^\top D w \neq \sum w_i d_i$. The weights are squared.

- **Connections**: W4 factor models. The Part 2 neutrality constraint sets $\beta_p = 0$, and Part 3 sets $B^\top a = 0$.

1.2 Mandates: the objective depends on the client [S6–S7]

- **Definition**: the three problems of the week share the same weight vector but penalise different mistakes.

Mandate	Objective	Penalises
Absolute return	$\max_w \ \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$	total variance
Factor neutral	$\max_w \ \alpha^\top w - \frac{\lambda}{2} w^\top D w$ s.t. $B^\top w = 0$	residual variance; forbids factor bets
Index tracking	$\min_w \ E[(w^\top \tilde{r} - \tilde{r}_b)^2]$ + trading costs	deviation from benchmark

- **Week 2 handoff [S6]**:
  - Already established: feasible set, efficient frontier (return $w^\top \mu$ vs risk $w^\top \Sigma w$), tangency portfolio, two-fund separation.
  - New questions: (i) μ, Σ carry estimation error; (ii) real mandates impose constraints; (iii) the objective may be benchmark tracking.
  - The lecture studies "the mapping from uncertain inputs and real mandates to implementable weights".

- ⚠ **[Beyond slides] second moment vs variance**: with $\tilde{r}_b = b^\top \tilde{r}$,

$$E[(w^\top \tilde{r} - \tilde{r}_b)^2] = E[(a^\top \tilde{r})^2] = a^\top \Sigma a + (a^\top \mu)^2 = \text{TE}^2 + (\text{expected active return})^2.$$

  - The [S7] objective is the *mean squared* tracking difference. TE [S41] is its *volatility* part.
  - They coincide when $a^\top \mu = 0$, and the difference is usually negligible for trackers.

- **Trap**: a factor-neutral mandate uses α and D, not μ and Σ. This is only legitimate *because* $B^\top w = 0$ (see 1.10).

1.3 Estimation risk and the "error maximizer" [S8–S10]

- **Definition**: MVO is an **error maximizer**. It places its biggest bets where return estimates are least reliable, so a small error in μ is amplified into a large position instead of averaging away (Michaud's term [Beyond slides: attribution]).

- **Formula**:

$$\max_w \ w^\top \mu - \frac{\lambda}{2} w^\top \Sigma w \ \Rightarrow \ w^* = \frac{1}{\lambda} \Sigma^{-1} \mu \ \Rightarrow \ \Delta w = \frac{1}{\lambda} \Sigma^{-1} \Delta \mu.$$

  - w^* is linear in μ, so the input error maps to a weight error through Σ^{-1}.

- **Intuition**:
  - Σ^{-1} has eigenvalues $1/d_j$, where d_j are the eigenvalues of Σ. The **lowest-variance directions** (e.g. a long–short spread between near-identical assets) get the biggest multipliers.
  - Statistical reading: the optimizer bets on estimation noise.
  - Portfolio reading: it builds long–short pairs to exploit tiny estimated return differences between similar assets.
- **Assumptions**: unconstrained quadratic problem, Σ invertible, residual cash earns R_f.
- **Derivation**:
  - FOC: $\mu - \lambda\Sigma w = 0$, so $w^* = \frac{1}{\lambda}\Sigma^{-1}\mu$.
  - Linearity gives Δw.
  - **2-asset closed form [Beyond slides, exam-useful]**: equal vols σ and correlation ρ give $\Sigma^{-1} = \frac{1}{\sigma^2(1-\rho^2)}\begin{pmatrix} 1 & -\rho \\ -\rho & 1 \end{pmatrix}$. So

$$\Delta w_1 = \frac{\Delta\mu_1}{\lambda\sigma^2(1-\rho^2)}, \qquad \Delta w_2 = \frac{-\rho\,\Delta\mu_1}{\lambda\sigma^2(1-\rho^2)}.$$

  - With equal means m: $w_1 = w_2 = \frac{m}{\lambda\sigma^2(1+\rho)}$.
- **Worked example (slide [S9])**:
  - Inputs: $\sigma = 20\%$, $\rho = 0.95$, $\lambda = 4$, so $\Sigma = \begin{pmatrix} 0.04 & 0.038 \\ 0.038 & 0.04 \end{pmatrix}$.
  - Base case $\mu = (8\%, 8\%)$: $w_i = \frac{0.08}{4(0.04)(1.95)} = \frac{0.08}{0.312} = 0.2564$, so $w^* = (0.256, 0.256)$.
  - Revise μ_1 to 9%: $1 - \rho^2 = 0.0975$, so $\Delta w_1 = \frac{0.01}{4(0.04)(0.0975)} = \frac{0.01}{0.0156} = 0.641$ and $\Delta w_2 = -0.95(0.641) = -0.609$.
  - New weights: $w^* = (0.897, -0.353)$.
  - **A 1-point forecast change moves asset 1 by 64 pp** (code [S10] prints 64.1 pp).
- ⚠ **Slide [S10] error**: "Try $\rho = 0.2$ and the same nudge moves the position by about 3 points instead of 64." The correct figure is $\Delta w_1 = \frac{0.01}{4(0.04)(0.96)} = 0.0651$, i.e. **≈6.5 pp** (verified numerically; $\Delta w_2 = -1.3$ pp). The qualitative point stands: the multiplier $1/(1-\rho^2)$ drops from 10.3 to 1.04.
- **Eigen view [Beyond slides]**:
  - Σ above has eigenvectors $(1,1)/\sqrt{2}$ with $d = 0.078$, and $(1,-1)/\sqrt{2}$ with $d = 0.002$.
  - Decompose the new mean: $\mu = 0.085(1,1) + 0.005(1,-1)$.
  - MVO divides each component by λd: $\frac{0.085}{0.312} = 0.272$ and $\frac{0.005}{0.008} = 0.625$.
  - Result: $w = 0.272(1,1) + 0.625(1,-1) = (0.897, -0.353)$.
  - A 0.5% spread in the $(1,-1)$ direction buys a 0.625 long–short position.
- **Interpretation / limits**:
  - $\rho \to 1$: $1 - \rho^2 \to 0$ and $\Delta w \to \infty$ (a singular Σ has an arbitrage-like spread).
  - $\rho = 0$: $\Delta w_1 = \Delta\mu_1/(\lambda\sigma^2)$ and $\Delta w_2 = 0$.
  - Larger λ scales all bets down proportionally but does **not** fix the *relative* instability.
- **Traps**:
  - Thinking the problem is "Σ is wrong". The amplification happens even with the true Σ; errors in μ matter most. (Chopra–Ziemba: mean errors ≈10× as costly as covariance errors [Beyond slides].)
  - Forgetting that Δw_2 moves opposite to Δw_1 when $\rho > 0$.
- **Connections**: W2 MVO and the tangency portfolio (direction $\Sigma^{-1}\mu$); estimator noise $\mathrm{SE}(\hat{\mu}) = \sigma/\sqrt{T}$ from W2. All three Part 1 fixes shrink this.

- **Implementation ([S10])**:

```
import numpy as np
s, rho, lam = 0.20, 0.95, 4.0
S = np.array([[s*s, rho*s*s], [rho*s*s, s*s]])
mvo = lambda mu, S, lam: np.linalg.solve(S, mu) / lam      # w = (1/lam) S^-1 mu
w0, w1 = mvo(np.array([.08, .08]), S, lam), mvo(np.array([.09, .08]), S, lam)
print(w0.round(3), w1.round(3), round(100*(w1[0]-w0[0]), 1), "pp")   # [0.256 0.256] [0.897
 ↪  -0.353] 64.1 pp
```

Pitfall: use `solve`, never `inv`. Check `np.linalg.cond(S)`; a large condition number means an unstable w.

1.4 The L2 norm [S13]

- **Definition**: $\|w\|_2 = \sqrt{w_1^2 + \cdots + w_n^2}$, the Euclidean length of the weight vector. The penalty uses the **squared** norm $\|w\|_2^2 = \sum_i w_i^2 = w^\top w$.
- **Example (slide)**: $w = (3, 4)$ gives $\|w\| = \sqrt{9+16} = 5$ and $\|w\|^2 = 25$.
- **Intuition**: the squared norm has no square root, so the penalty grows **quadratically** and punishes big weights hardest.
- **Related [Beyond slides]**:
  - $\|w\|_1 = \sum |w_i|$ is the gross exposure / leverage. It is used for turnover in [S23].
  - $\|w\|_0 = \#\{i : w_i \neq 0\}$ is not a true norm. It is used for cardinality in [S45].
- **Trap**: "$\|w\|^2$" in the slides means the squared L2 norm, not the L2 norm of something called w^2.

1.5 Ridge (L2) regularisation [S11–S18]

- **Definition**: add a size penalty $\frac{\eta}{2}\|w\|_2^2$ to the MVO objective:

$$\max_w \ \hat{\mu}^\top w - \frac{\lambda}{2} w^\top \hat{\Sigma} w - \frac{\eta}{2}\|w\|_2^2 \quad \Rightarrow \quad w^* = (\lambda\hat{\Sigma} + \eta I)^{-1}\hat{\mu}.$$

- **Intuition**:
  - Adding η "bounds every direction in the inverse". The eigenvalues of $\lambda\hat{\Sigma} + \eta I$ are $\lambda d_j + \eta \geq \eta$, so the lowest-variance directions can no longer explode.
  - η "trades a little in-sample fit for less sensitivity to noisy inputs" [S14].
  - The result is modest, stable weights instead of extreme long–short bets [S12].
- **Symbols**: $\eta \geq 0$ in return units, $I = N \times N$ identity, and $\hat{\cdot}$ = sample estimate.
- **Assumptions**: unconstrained (no budget). η is chosen **out of sample** (e.g. by cross-validation or a rolling backtest), "not on the fit" [S18]. There is no "right" value.
- **Derivation ([S15], exam-reproducible)**:
  1. $f(w) = \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w - \frac{\eta}{2} w^\top w$.
  2. $\nabla f = \mu - \lambda\Sigma w - \eta w$.
  3. Set it to 0: $(\lambda\Sigma + \eta I)w = \mu$.
  4. $w^* = (\lambda\Sigma + \eta I)^{-1}\mu$.

  - This is a maximum because the Hessian $-(\lambda\Sigma + \eta I)$ is negative definite.
- **Equivalent views [Beyond slides]**:
  - **Covariance shrinkage**: $w^* = \frac{1}{\lambda}(\Sigma + \frac{\eta}{\lambda} I)^{-1}\mu$, i.e. MVO with Σ shrunk toward a scaled identity (Ledoit–Wolf flavour).
  - **Eigen-shrinkage**: $w^* = \sum_j \frac{q_j^\top \mu}{\lambda d_j + \eta} q_j$. Relative to MVO, direction j is scaled by $\frac{\lambda d_j}{\lambda d_j + \eta}$, so **low-variance directions are shrunk most**.

  - **Ridge regression** analogy: $(X^\top X + \eta I)^{-1} X^\top y$.

- **Intuition slides**:
  - **[S16] Spreads bets**: with the same total exposure, one weight of 1.0 has $\|w\|^2 = 1$, while four weights of 0.25 have $\|w\|^2 = 4(0.0625) = 0.25$. The diversified book costs **4× less** penalty.
  - **[S17] No signal gives 1/N**: $\min \|w\|^2$ s.t. $\mathbf{1}^\top w = 1$ gives $w_i = 1/N$.
    * Proof [Beyond slides]: $\mathcal{L} = w^\top w - \gamma(\mathbf{1}^\top w - 1)$, so $2w = \gamma\mathbf{1}$, $w = \frac{\gamma}{2}\mathbf{1}$, and the budget gives $\gamma = 2/N$.
    * Minimum value is $1/N$.
    * So the L2 penalty is "pure diversification".

- **Worked example [Beyond slides]** (the slide's 2-asset case, $\mu = (9\%, 8\%)$, $\lambda = 4$, $\eta = 0.05$):
  - $\lambda\Sigma + \eta I = \begin{pmatrix} 0.21 & 0.152 \\ 0.152 & 0.21 \end{pmatrix}$, with determinant $0.0441 - 0.023104 = 0.020996$.
  - $w_1 = \frac{0.21(0.09) - 0.152(0.08)}{0.020996} = \frac{0.0189 - 0.01216}{0.020996} = 0.3210$.
  - $w_2 = \frac{0.21(0.08) - 0.152(0.09)}{0.020996} = \frac{0.0168 - 0.01368}{0.020996} = 0.1486$.
  - $w^* = (0.321, 0.149)$ **vs MVO** $(0.897, -0.353)$**. The short disappears.**
  - Eigen check: $\frac{0.085}{0.362} = 0.2348$ and $\frac{0.005}{0.058} = 0.0862$, so $w = (0.321, 0.149)$ ✓. The spread direction is shrunk by $0.008/0.058 = 0.14$; the common direction only by $0.312/0.362 = 0.86$.
  - For $\eta = 0.2$: $(0.190, 0.142)$. For $\eta = 1$: $(0.070, 0.060)$.

- **Slide code output [S18]** (10-asset dataset `MU`, `SIG`; the data are not printed in the slides, so it cannot be reproduced by hand):

η	gross $\sum \|w_i\|$	max $\|w_i\|$	short
0.00	13.42	3.24	4.99
0.05	3.95	0.71	0.99
0.20	1.71	0.25	0.23
1.00	0.58	0.10	0.00

- At $\eta = 0$ the book has 13.4× leverage and a 499% short. At $\eta = 1$ it is 0.58× with no shorts.
- **Interpretation / limits**:
  - $\eta \to 0$ gives plain MVO.
  - $\eta \to \infty$ gives $w^* \approx \mu/\eta \to 0$. **Without** a budget constraint the book shrinks to cash and becomes proportional to μ, *not* 1/N.
  - 1/N appears only with $\mathbf{1}^\top w = 1$ and no signal [S17].
  - Gross leverage falls monotonically in η.
- **Traps**:
  - Writing $(\lambda\Sigma + \eta)^{-1}$ without I.
  - Dividing by λ again. The ridge formula has **no** $1/\lambda$ outside; it's inside the matrix.
  - Tuning η in-sample (in-sample fit always prefers $\eta = 0$).
  - Expecting ridge to set weights exactly to zero. It doesn't; that's L1/lasso.
- **Connections**: W2 MVO; robust optimisation (1.6, also shrinkage); turnover reduction (1.7).
- **Implementation ([S18])**:

```python
def ridge(mu, S, lam, eta):
    return np.linalg.solve(lam*S + eta*np.eye(len(mu)), mu)   # (lam S + eta I)^-1 mu
for eta in [0, .05, .2, 1]:
    print(eta, ridge(np.array([.09, .08]), S, 4.0, eta).round(3))   # 0.05 -> [0.321 0.149]
```

Pitfall: η is scale-dependent. If returns are monthly rather than annual, the same η means a different amount of shrinkage.

1.6 Robust optimisation (ellipsoidal mean uncertainty) [S19–S21]

- **Definition**:
  - Treat the mean as uncertain, lying in an ellipsoid around the estimate:

$$U = \{\hat{\mu} + S^{1/2}u : \|u\|_2 \le \varepsilon\} \iff U = \{\tilde{\mu} : (\tilde{\mu} - \hat{\mu})^\top S^{-1}(\tilde{\mu} - \hat{\mu}) \le \kappa^2\},\ \kappa \equiv \varepsilon.$$

  - Then choose w to win under the **worst** mean in U:

$$\max_w \min_{\tilde{\mu} \in U} \left(\tilde{\mu}^\top w - \frac{\lambda}{2} w^\top \Sigma w\right).$$

- **Intuition**:
  - "Hedge the forecast directly": pick the portfolio that is best even at the worst mean.
  - The worst-case $\tilde{\mu}$ sits on the ellipse boundary, pushed *against* the direction of w (picture [S20]).
  - The more you bet, the more the adversary hurts you.
- **Closed form (slide [S21], with $S = \Sigma$)**:

$$w^* = w_0 \cdot \max\left(0,\ 1 - \frac{\kappa}{t}\right), \qquad w_0 = \frac{1}{\lambda}\Sigma^{-1}\hat{\mu}, \quad t = \sqrt{\hat{\mu}^\top \Sigma^{-1}\hat{\mu}}.$$

  - κ is how much you distrust μ. t is the size of the forecast, which equals the **maximum attainable Sharpe ratio** if $\hat{\mu}$ is an excess return [Beyond slides].
  - Once $\kappa \ge t$, you stop betting.
- **Derivation [Beyond slides, the slide only states the result]**:
  1. **Inner min**:

$$\min_{\|u\| \le \varepsilon} (\hat{\mu} + S^{1/2}u)^\top w = \hat{\mu}^\top w - \varepsilon \|S^{1/2}w\|_2 = \hat{\mu}^\top w - \varepsilon\sqrt{w^\top S w}$$

     - By Cauchy–Schwarz, the minimiser is $u^* = -\varepsilon S^{1/2}w / \|S^{1/2}w\|$.
     - So the worst mean is $\tilde{\mu}^* = \hat{\mu} - \varepsilon\, Sw/\sqrt{w^\top S w}$.
  2. **Robust counterpart**: $\max_w\ \hat{\mu}^\top w - \kappa\sqrt{w^\top S w} - \frac{\lambda}{2} w^\top \Sigma w$. This is a "mean minus κ·stdev" penalty, i.e. concave (SOCP).
  3. **With** $S = \Sigma$, the FOC ($w \ne 0$) is $\hat{\mu} - \kappa\frac{\Sigma w}{\sigma_p} - \lambda\Sigma w = 0$, where $\sigma_p = \sqrt{w^\top \Sigma w}$. So $w \propto \Sigma^{-1}\hat{\mu}$.
  4. Write $w = c\,\Sigma^{-1}\hat{\mu}$, so $\sigma_p = ct$. Substituting gives $c\lambda + \kappa/t = 1$, so $c = \frac{1}{\lambda}(1 - \kappa/t)$.
  5. If $\kappa \ge t$, the subgradient condition at $w = 0$ holds, so $w^* = 0$.
- **Worked example [Beyond slides]** (slide 2-asset case, $\hat{\mu} = (9\%, 8\%)$, $\lambda = 4$):
  - $\Sigma^{-1}\hat{\mu} = \lambda w_0 = 4(0.8974, -0.3526) = (3.590, -1.410)$.
  - $t^2 = 0.09(3.590) + 0.08(-1.410) = 0.3231 - 0.1128 = 0.2103$, so $t = 0.4585$.
  - With $\kappa = 0.2$: shrink factor $1 - 0.2/0.4585 = 0.5638$, so $w^* = (0.506, -0.199)$ (verified by numerical optimisation).
  - $\kappa = 0.1$ gives $(0.702, -0.276)$.
  - Any $\kappa \ge 0.4585$ gives $w^* = 0$.
- **Slide code output [S21]** (10-asset data):

κ	gross $\sum \|w_i\|$	expected $\mu^\top w$
0.0	13.42	58.92%
0.5	9.05	39.73%
1.0	4.68	20.54%

- Consistency check [Beyond slides]: $1 - 0.5/t = 39.73/58.92$ implies $t \approx 1.535$.

- Then $\mu^\top w_0 = t^2/\lambda = 2.357/4 = 58.9\%$ ✓, and at $\kappa = 1$ the factor is 0.349, giving $13.42 \times 0.349 = 4.68$ ✓.
- **Interpretation / limits**:
  - $\kappa = 0$ gives MVO.
  - The shrink factor is linear in κ.
  - t large (a strong forecast relative to risk) makes it hard to wipe out.
  - Expected return $\hat{\mu}^\top w^* = \frac{t^2}{\lambda}(1 - \kappa/t)$.
- **Ridge vs robust ([S21] footer)**:
  - Ridge shrinks every weight toward zero, and shrinks low-variance directions *more*, so it changes the **shape** of w.
  - Robust (with $S = \Sigma$) shrinks the whole tilt along **one direction**: the proportions of w_0 are unchanged, only the scale.
  - They look similar in the leverage table; they are not the same in general (and with $S \neq \Sigma$ robust also changes shape).
- ⚠ **Units of κ**: the code comment says "κ is … measured in standard errors", but the code passes $S =$ `SIG` (the return covariance), so κ is measured in units of return standard deviation.
  - If U is built from the **estimation error** covariance $S = \Sigma/T$ (T observations), then $\kappa\sqrt{w^\top S w} = \frac{\kappa}{\sqrt{T}}\sigma_p$.
  - The shrink factor then becomes $1 - \frac{\kappa}{\sqrt{T}\,t}$.
  - State which S you use.
- **Traps**:
  - Forgetting the $\max(0, \cdot)$ (weights never flip sign under this robustification).
  - Plugging $t = \hat{\mu}^\top \Sigma^{-1} \hat{\mu}$ without the square root.
  - Believing robust = ridge.
- **Connections**: W2 Sharpe ratio / tangency direction $\Sigma^{-1}\mu$. Also a form of shrinkage estimation (like Bayes–Stein / Black–Litterman toward zero tilt [Beyond slides]).
- **Implementation ([S21])**:

```python
def robust(mu, S, lam, kappa):
    w0 = np.linalg.solve(S, mu) / lam            # nominal tilt
    t  = np.sqrt(mu @ np.linalg.solve(S, mu))    # size of forecast (max Sharpe if S = Sigma)
    return w0 * max(0.0, 1.0 - kappa / t)
print(robust(np.array([.09, .08]), S, 4.0, 0.2).round(3))   # [ 0.506 -0.199]
```

Pitfall: the closed form holds only for $S \propto \Sigma$. For a general S, solve the SOCP numerically (e.g. cvxpy).

1.7 Trading costs and the L1 turnover penalty [S22–S24]

- **Definition**: a manager trades from current **drifted** holdings w^-, not a fresh portfolio each month:

$$\max_w \ \hat{\mu}^\top w - \frac{\lambda}{2} w^\top \hat{\Sigma} w - \tau \|w - w^-\|_1, \qquad \|w - w^-\|_1 = \sum_i |w_i - w_i^-|.$$

- **Intuition**:
  - The best *target* weight may be a poor *trade*.
  - The L1 term creates a **no-trade region**: a small forecast change need not trigger a trade.
  - τ can reflect fees plus estimated market impact.
- **Symbols**: τ = cost per unit of notional traded, as a fraction of NAV (20 bp = 0.002).
- **Why L1, not L2 [Beyond slides]**:
  - $|x|$ has a **kink** at 0, so the marginal cost of the first dollar traded is $\tau > 0$. Small benefits don't overcome it, which gives exact zeros (no trade).
  - A quadratic cost has zero marginal cost at 0, so it always trades a little.

- **No-trade region derivation [Beyond slides]** (one risky asset: variance σ^2, target $w^\circ = \mu/(\lambda\sigma^2)$):
  - Subgradient FOC: $\mu - \lambda\sigma^2 w - \tau s = 0$, where $s = \text{sign}(w - w^-)$, and $s \in [-1, 1]$ if $w = w^-$.
  - Keep w^- iff $|\mu - \lambda\sigma^2 w^-| \le \tau$, i.e.

$$w^- \in \left[\frac{\mu - \tau}{\lambda\sigma^2},\ \frac{\mu + \tau}{\lambda\sigma^2}\right].$$

  - Otherwise trade only **to the nearest edge** of the band, not to the target.
  - Result: $w^* = \text{clip}(w^-,\ \text{lo},\ \text{hi})$.
- **Worked example [Beyond slides]**:
  - Inputs: $\mu = 6\%$, $\sigma = 20\%$, $\lambda = 4$, so $\lambda\sigma^2 = 0.16$ and target = 0.375. $\tau = 50$ bp.
  - Band = [0.055/0.16, 0.065/0.16] = [0.34375, 0.40625].
  - If $w^- = 0.30$: **buy to 0.344**, not to 0.375.
  - If $w^- = 0.38$: **no trade**.
  - If $w^- = 0.50$: sell to 0.406.
- **Backtests must charge ([S23])**:
  - Use post-return drifted weights, $w_i^- = \frac{w_i(1+r_i)}{1+w^\top r}$ [formula: Beyond slides].
  - Use actual reconstitution dates.
  - Use a cost model calibrated to traded notional.
  - "A backtest that does not charge for turnover will always prefer the most unstable strategy. Charge first, then rank." [S24]
- **Slide code [S24]**:
  - Turnover = $\sum |w^{new} - w^{old}|$ and cost = turnover × bps/10^4.
  - Moving from $1/N$ to the unconstrained MVO target: **turnover 13.22× NAV**.
  - At 20 bp that costs $13.22 \times 0.002 =$ **2.64%**, so a gross 1.20% month becomes **net −1.44%**. Cost exceeds the month's gross return.
  - All three Part 1 fixes cut turnover as a side effect.
- ⚠ **Turnover convention**: the code comment says "one-way", but $\sum_i |\Delta w_i|$ counts buys **and** sells.
  - Industry "one-way turnover" is usually $\frac{1}{2}\sum |\Delta w_i|$ (for a fully invested long-only book, buys = sells).
  - Charging bps × $\sum |\Delta w_i|$ is correct if the bps is **per unit traded**. If a cost is quoted "per unit of one-way turnover", halve the base or double the rate.
  - Always state the convention.
- **Worked example [Beyond slides]**:
  - Trade from $w^{old} = (0.25, 0.25, 0.25, 0.25)$ to $w^{new} = (0.40, 0.30, 0.20, 0.10)$.
  - $\sum |\Delta| = 0.15 + 0.05 + 0.05 + 0.15 = 0.40$, so one-way turnover = 0.20.
  - Cost at 20 bp per unit traded = $0.40 \times 0.002 = 0.08\%$.
  - Gross 1.00% becomes **net 0.92%**.
- **Interpretation / limits**:
  - $\tau \to 0$: always trade to target.
  - $\tau \ge |\mu - \lambda\sigma^2 w^-|$: never trade.
  - A larger $\lambda\sigma^2$ narrows the band (risk makes being off-target costly).
- **Traps**:
  - Rebalancing from last period's *target* instead of the *drifted* weights.
  - Applying costs to gross (not net) weights.
  - bps confusion: 20 bp = 0.20% = 0.002.
  - Comparing strategies gross of costs.
- **Connections**: W2 backtesting; reappears in index tracking [S45].

- **Implementation**:

```
def net_of_cost(w_new, w_old, gross_ret, bps=20):
    turnover = np.abs(w_new - w_old).sum()       # sum of |trades| (buys + sells)
    cost = turnover * bps / 1e4
    return gross_ret - cost, turnover, cost
def drift(w, r):                                  # post-return weights before rebalancing
    g = w * (1 + r); return g / g.sum()
def no_trade(mu, s2, lam, tau, w_prev):           # 1-asset L1 solution
    return float(np.clip(w_prev, (mu - tau)/(lam*s2), (mu + tau)/(lam*s2)))
print(no_trade(.06, .04, 4, .005, .30))           # 0.34375
```

Pitfall: for many assets the L1 problem has no closed form. Use a convex solver, or split $w - w^- = p - m$ with $p, m \geq 0$ so it becomes a smooth QP.

1.8 Return decomposition: skill, factor, luck [S25–S26]

- **Definition**: every asset's return is

$$\tilde{r} = \underbrace{\alpha}_{\text{skill}} + \underbrace{B\tilde{f}}_{\text{factor}} + \underbrace{\tilde{\varepsilon}}_{\text{luck}}.$$

  - α is your forecast of what the factors don't explain.
  - $B\tilde{f}$ is what market and style factors drive.
  - $\tilde{\varepsilon}$ is idiosyncratic, with $E[\tilde{\varepsilon}] = 0$.

- **Intuition**:

  - Part 2 uses the factor model to separate the alpha you *meant* to take from the factor exposure you *didn't mean* to take, which can be hedged.
  - Factor neutrality targets the **middle piece** by forcing $B^\top w = 0$.

- **Symbols**: $\tilde{r}, \alpha, \tilde{\varepsilon}$ are $N \times 1$; B is $N \times K$; $\tilde{f}$ is $K \times 1$.

- ⚠ **Notation vs W4**: W4 writes $\tilde{R}_i - R_f = \alpha_i + \sum_k \beta_{ik}\tilde{F}_k + \tilde{\varepsilon}_i$.

  - Week 5's $\tilde{r}$ plays the role of the **excess** return, $\tilde{f} = \tilde{F}$, and row i of $B = (\beta_{i1}, \ldots, \beta_{iK})$.
  - With $\tilde{f}$ in raw (non-demeaned) form, $E[\tilde{r}] = \alpha + B\mu_f$. This is exactly [S29]'s $\mu = B\lambda + \alpha$.

- **Traps**:

  - "Alpha" here is a **forecast** (a model input), not a realised regression intercept, although the intercept is how it is estimated [S29].
  - $\tilde{\varepsilon}$ (a shock) is not the robust-set radius ε.

- **Connections**: W3 CAPM ($K = 1$, the market factor); W4 multi-factor models, including the Aug-2007 quant crisis, where funds shared an unrecognised factor bet. That episode is exactly what neutrality is designed to prevent.

1.9 Beta-neutral portfolios and the cost of neutrality [S27–S28]

- **Definition**: the portfolio factor exposure is

$$B^\top w = \sum_i w_i \beta_i \quad (K \times 1;\ \beta_i = \text{row } i \text{ of } B).$$

  - **Beta-neutral** $\Leftrightarrow B^\top w = 0$: every factor's net exposure is zero.

- **Intuition ([S27] picture)**:

  - Each holding contributes weight × beta. Two long contributions (+0.60, +0.24) are offset by a short (−0.84), so the net is 0.
  - The factor moves the longs and shorts by equal and opposite amounts.

- **Cost of neutrality [S28]**:
  - With $B^\top w = 0$: $E[w^\top \tilde{r}] = w^\top \alpha + (B^\top w)^\top \mu_f = w^\top \alpha$. **Only your alpha survives, with no factor premium.**
  - The market pays a premium for bearing factor risk. Refuse the risk ($\beta = 0$) and you refuse the premium.
  - "A strategy cannot claim both zero beta and the beta premium; neutrality is a choice to be paid only for skill."
  - Picture: portfolio return vs factor return is a sloped line for $\beta > 0$ and a flat line for $\beta = 0$.
- **Worked example [Beyond slides]**:
  - One factor with market premium $\mu_f = 6\%$, and two stocks with $\beta = (1.2, 0.8)$ and $\alpha = (1\%, -0.5\%)$.
  - $w = (0.5, -0.5)$ gives $\beta_p = 0.2$ and $E[r_p] = 0.75\% + 0.2(6\%) = 1.95\%$.
  - The neutral version $(0.412, -0.618)$ (see 1.12) earns $E[r_p] = 0.721\% + 0 = \mathbf{0.721\%}$. Most of the drop is the forgone premium; only 0.029% is lost alpha.
- **Interpretation**:
  - $\beta_p > 0$ means the book earns $\beta_p \mu_f$ on average but carries $\beta_p^2 \Omega$ of factor variance.
  - Neutrality removes both. What remains is the alpha and $w^\top D w$.
- **Traps**:
  - Beta-neutral ≠ dollar-neutral. $\mathbf{1}^\top w = 0$ is a different constraint; $w = (0.5, -0.5)$ is dollar-neutral but has $\beta_p = 0.2$.
  - Neutral to the market factor ≠ neutral to style factors. Each column of B needs its own zero.
  - Claiming "market-neutral alpha" while collecting a factor premium.
- **Connections**: W3 (α vs β), W4 (factor premia), Part 3 relative neutrality.

1.10 Residual space and the factor covariance $\Sigma = B\Omega B^\top + D$ [S29–S30]

- **Definition**: neutrality strips the factor part from both the return and the risk.

$$\mu = B\lambda + \alpha \;\Rightarrow\; \alpha = \mu - B\lambda \qquad (\mu \to \alpha)$$

$$\Sigma = B\Omega B^\top + D \;\Rightarrow\; D = \Sigma - B\Omega B^\top \qquad (\Sigma \to D)$$

  - Here $\lambda = K \times 1$ factor risk premia ($= E[\tilde{f}]$), **not** risk aversion ⚠.
- **Estimation (slides)**:
  - α: the intercept when returns are regressed on the factors. It is the residual mean, i.e. pure selection skill.
  - B: regress each stock's returns on the factor returns. The slopes are that stock's row of B.
  - Ω: sample covariance of the factor returns, $K \times K$.
  - D: variance of each stock's regression residuals, placed on the diagonal.
- **Derivation ([S30], optional on slide, exam-worthy)**:

$$\tilde{r} = B\tilde{f} + \tilde{\varepsilon} \Rightarrow \Sigma = \mathrm{Cov}(B\tilde{f} + \tilde{\varepsilon}) = B\,\mathrm{Cov}(\tilde{f})B^\top + \mathrm{Cov}(\tilde{\varepsilon}) + B\,\mathrm{Cov}(\tilde{f}, \tilde{\varepsilon}) + \mathrm{Cov}(\tilde{\varepsilon}, \tilde{f})B^\top.$$

  - (Constants such as α don't affect covariance.)
  - Assumption 1: shocks are uncorrelated with factors, so the cross terms vanish.
  - Assumption 2: shocks are uncorrelated across stocks, so D is **diagonal** (one specific variance per stock).
  - Result: $\Sigma = B\Omega B^\top + D$.
- **Why the pieces drop out under neutrality ([S29] footer; algebra is Beyond slides)**. If $B^\top w = 0$:

  - Return: $\mu^\top w = (B\lambda + \alpha)^\top w = \lambda^\top (B^\top w) + \alpha^\top w = \alpha^\top w$.
  - Risk: $w^\top \Sigma w = (B^\top w)^\top \Omega (B^\top w) + w^\top D w = w^\top D w$.
  - So standard MVO $\max \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$ **restricted to** $B^\top w = 0$ is *identical* to $\max \alpha^\top w - \frac{\lambda}{2} w^\top D w$ s.t. $B^\top w = 0$.
- **Benefits of the structure [Beyond slides]**:
  - Parameters drop from $N(N+1)/2$ to $NK + K(K+1)/2 + N$. For $N = 500$ and $K = 5$: 125,250 vs 3,015.
  - Σ is positive definite if $D \succ 0$.
  - D^{-1} is trivial to compute $(1/d_i)$.
- **Worked example**:
  - One factor with $\Omega = 0.04$, $B = (1.2, 0.8)^\top$, $D = \mathrm{diag}(0.02, 0.01)$.
  - $\Sigma = 0.04 \begin{pmatrix} 1.44 & 0.96 \\ 0.96 & 0.64 \end{pmatrix} + D = \begin{pmatrix} 0.0776 & 0.0384 \\ 0.0384 & 0.0356 \end{pmatrix}$.
  - Vols are 27.9% and 18.9%, and the implied correlation is $0.0384/\sqrt{0.0776 \times 0.0356} = 0.73$.
- **Traps**:
  - Assuming D is diagonal when an omitted factor (e.g. industry) makes residuals correlated. Then $w^\top D w$ understates risk.
  - The regression must use the same return definition (excess vs raw) as the model.
  - α from a regression is a *historical* intercept. Using it as a forecast is a modelling choice.
- **Connections**: W4 time-series regressions for β; W3 Jensen's alpha.
- **Implementation (homework step 1 [S48])**:

```
# R: T x N excess returns, F: T x K factor excess returns
X = np.column_stack([np.ones(len(F)), F])
coef, *_ = np.linalg.lstsq(X, R, rcond=None)     # (K+1) x N
alpha, B = coef[0], coef[1:].T                     # alpha: N,  B: N x K
resid = R - X @ coef
D = np.diag(resid.var(axis=0, ddof=X.shape[1]))   # specific variances
Omega = np.atleast_2d(np.cov(F, rowvar=False))
Sigma = B @ Omega @ B.T + D
```

Pitfall: `ddof` should subtract the number of fitted parameters ($K+1$). Use `np.atleast_2d` so the $K = 1$ case keeps the right shape.

1.11 The factor-neutral optimisation and its closed form [S31–S32]

- **Problem [S31]**:

$$\max_w \ \alpha^\top w - \frac{\lambda}{2} w^\top D w \quad \text{s.t.} \quad B^\top w = 0.$$

  - $\alpha^\top w$ = expected alpha.
  - $\frac{\lambda}{2} w^\top D w$ = residual risk you pay to carry.
  - $B^\top w = 0$ = zero net exposure to every factor (K linear constraints).
- **Derivation ([S32], reproduce in the exam)**:
  1. Lagrangian: $\mathcal{L}(w, \nu) = \alpha^\top w - \frac{\lambda}{2} w^\top D w - \nu^\top B^\top w$, with $\nu \in \mathbb{R}^K$.
  2. FOC: $\alpha - \lambda D w - B\nu = 0$, so $w = \frac{1}{\lambda} D^{-1}(\alpha - B\nu)$.
  3. Impose $B^\top w = 0$: $B^\top D^{-1} \alpha = B^\top D^{-1} B \, \nu$, so

$$\nu = (B^\top D^{-1} B)^{-1} B^\top D^{-1} \alpha.$$

     - λ cancels.
     - [Beyond slides] ν is the **GLS regression coefficient of α on B** (weights D^{-1}), i.e. the factor-explained part of your alpha.

4. Neutral weights:

$$w^* = \frac{1}{\lambda}\left[D^{-1} - D^{-1}B(B^\top D^{-1}B)^{-1}B^\top D^{-1}\right]\alpha.$$

- **Compare [S32 side box]**: unconstrained MVO $w^* = \frac{1}{\lambda}\Sigma^{-1}\mu$. Neutral MVO replaces $\Sigma \to D$ and $\mu \to \alpha$, then removes the factor-aligned part.
- **Assumptions**:
  - $D \succ 0$ (invertible).
  - B has full column rank K, so $B^\top D^{-1}B$ is invertible.
  - Need $N > K$; otherwise the only neutral portfolio is $w = 0$.
  - No budget constraint (the book is self-financing; see the traps).
- **Useful identities [Beyond slides]**:
  - $\alpha^\top w^* = \lambda\, w^{*\top} D w^*$ (since $w^{*\top} B\nu = 0$). So the optimal utility is $\frac{1}{2}\alpha^\top w^*$.
  - If $\alpha = Bc$ for some c (alpha is *entirely* explained by factor loadings), then $\nu = c$ and $w^* = 0$: there is nothing left to earn once neutral.
- **Interpretation**:
  - λ scales the book. ν measures how much of the "alpha" was really factor tilt.
  - With $K = 1$: $w^* = \frac{1}{\lambda}D^{-1}(\alpha - \nu B)$, i.e. hold each stock in proportion to its **beta-adjusted alpha** $(\alpha_i - \nu\beta_i)/d_i$.
- **Traps**:
  - Using Σ instead of D in the objective. That's still valid under neutrality, but then the formula has Σ^{-1} and the projection is in the Σ metric. The result is the same w^* only because $B^\top w = 0$ makes the factor terms vanish.
  - Dropping the transpose: $B^\top D^{-1}B$ is $K \times K$, while $BD^{-1}B^\top$ is the wrong shape.
  - Adding $\mathbf{1}^\top w = 1$ when a neutral book is typically **zero-investment** (dollar- or beta-neutral long–short). If you need it, add it as an extra row in the constraint matrix.

1.12 Constrained optimum as a projection [S33–S36]

- **Statement ([S33], plain MVO motivating example)**:
  1. The **ideal**: $w_0 = \frac{1}{\lambda}\Sigma^{-1}\mu$ is the centre of the risk-ellipse contours.
  2. On the feasible set, maximising return − risk is the same as minimising the **risk-distance** to w_0.
  3. The nearest feasible point is where a risk contour is tangent to the constraint.
  - So constrained optimisation = projecting the ideal solution onto the feasible set, with distance measured in risk.
- **Why [Beyond slides]**:
  - $\mu = \lambda\Sigma w_0$, so $\mu^\top w - \frac{\lambda}{2}w^\top\Sigma w = -\frac{\lambda}{2}(w - w_0)^\top\Sigma(w - w_0) + \frac{\lambda}{2}w_0^\top\Sigma w_0$.
  - Maximising the left side ⇔ minimising $\|w - w_0\|_\Sigma^2$.
- **Neutralising is a projection [S34]**:

$$w_0 = \frac{D^{-1}\alpha}{\lambda}, \qquad w^* = w_0 - D^{-1}B(B^\top D^{-1}B)^{-1}B^\top w_0.$$

  - Start from the ideal alpha portfolio and subtract the part pointing along the factors. The result is the closest neutral portfolio to w_0 in the D-metric.
  - Equivalent to [S32] because $B^\top w_0 = \frac{1}{\lambda}B^\top D^{-1}\alpha$.
  - $P = I - D^{-1}B(B^\top D^{-1}B)^{-1}B^\top$ is an (oblique) projector: $P^2 = P$ and $B^\top P = 0$ [Beyond slides].

- **General linear constraints [Beyond slides]**: $A^\top w = c$ in metric M gives

$$w^* = w_0 - M^{-1}A(A^\top M^{-1}A)^{-1}(A^\top w_0 - c).$$

  - Relative neutrality ($B^\top w = B^\top b$) uses $A = B$, $c = B^\top b$, $M = D$.

- **Worked example (slides [S35]-[S36])**:
  - Inputs: $B = (1.2, 0.8)^\top$, $\alpha = (1\%, -0.5\%)$, $D = \mathrm{diag}(0.02, 0.01)$, $\lambda = 1$.

  1. $w_0 = D^{-1}\alpha = (0.01/0.02,\ -0.005/0.01) = (0.500, -0.500)$.
  2. $B^\top w_0 = 1.2(0.5) + 0.8(-0.5) = 0.200$.
  3. $B^\top D^{-1} B = \frac{1.2^2}{0.02} + \frac{0.8^2}{0.01} = 72 + 64 = 136$.
  4. $D^{-1}B = (60, 80)$, scale $= 0.200/136 = 0.0014706$, so $w^* = (0.5 - 0.0882,\ -0.5 - 0.1176) =$ **(0.412, −0.618)**.
  5. Check: $B^\top w^* = 1.2(0.4118) + 0.8(-0.6176) = 0.4941 - 0.4941 = 0$ ✓.

  - Alpha: $\alpha^\top w_0 = 0.005 + 0.0025 = 0.750\%$ becomes $\alpha^\top w^* = 0.004118 + 0.003088 =$ **0.721%**. The **price of neutrality** is 0.029%.
  - [Beyond slides] Utility $\alpha^\top w - \frac{1}{2} w^\top D w$ goes from 0.375% to 0.360%, and $\alpha^\top w^* = w^{*\top} D w^* = 0.721\%$ ✓ (identity from 1.11 with $\lambda = 1$).
  - Code prints $B^\top w^* = -8.3 \times 10^{-17}$. That is zero to machine precision because the projection makes it zero **by construction, not by search**.

- **Interpretation**:
  - The projection moved *both* weights. Neutralising a +0.2 beta tilt required a larger short in the lower-residual-variance stock (the $D^{-1}B$ direction: stock 2 has $d = 0.01$, so it is cheap to adjust).
  - The scale $B^\top w_0 / (B^\top D^{-1} B)$ is how much to strip out.

- **Traps**:
  - Projecting with the Euclidean metric ($w_0 - B(B^\top B)^{-1} B^\top w_0$). That is neutral but **not** optimal.
  - Forgetting to divide by λ in w_0.

- **Implementation ([S36])**:

```
B = np.array([1.2, 0.8]); a = np.array([0.010, -0.005]); D = np.diag([0.02, 0.01]); lam = 1.0
Di = np.linalg.inv(D)                          # D diagonal: cheap
w0 = Di @ a / lam                              # ideal, ignoring the rule
w = w0 - (B @ w0) / (B @ Di @ B) * (Di @ B)   # project onto B'w = 0   (K = 1 form)
print(w.round(4), B @ w, 100*a @ w0, 100*a @ w)   # [0.4118 -0.6176] ~0 0.75 0.7206
# K >= 1 general form:
def neutral(alpha, B, D, lam):
    Di = np.diag(1/np.diag(D)); w0 = Di @ alpha / lam
    return w0 - Di @ B @ np.linalg.solve(B.T @ Di @ B, B.T @ w0)
```

Pitfall: in the $K = 1$ code, `B` is 1-D, so `B @ Di @ B` is a scalar. For $K > 1$, `B` must be $N \times K$ and you need `solve`.

1.13 Feasibility: long-only cannot be factor-neutral [S37]

- **Statement**: if every asset has $\beta_i > 0$, and long-only means $w_i \geq 0$ with $\mathbf{1}^\top w = 1$, then $\beta^\top w \geq \min_i \beta_i > 0$. **Zero beta needs shorts**, so the problem is infeasible.

- **Fix: relative neutrality**:

$$B^\top(w - b) = 0 \quad \Longleftrightarrow \quad B^\top w = B^\top b.$$

  - Don't zero the exposure; **match the benchmark's**. The book carries the benchmark's factor exposure by design, and you control deviations from it.

- **Proof [Beyond slides]**: $\beta^\top w = \sum w_i \beta_i \geq \min_i \beta_i \sum w_i = \min_i \beta_i > 0$. The long-only budget set is the simplex, and $\beta^\top w$ on it ranges over $[\min \beta_i, \max \beta_i]$.

- **Worked example [Beyond slides]**:
  - $\beta = (0.8, 1.1, 1.3)$, long-only: achievable betas lie in $[0.8, 1.3]$, so 0 is infeasible.
  - An equal-weight benchmark has $\beta_b = 1.067$, which is achievable (e.g. $w = b$, or $w = (0.4, 0.2, 0.4)$ with $0.32 + 0.22 + 0.52 = 1.06$, close).
- **Feasibility for general targets [Beyond slides]**: a target $\beta^\star$ is long-only-feasible iff $\min_i \beta_i \le \beta^\star \le \max_i \beta_i$ (one factor). With K factors, the target must lie in the convex hull of the rows of B.
- **Traps**:
  - Solving a neutral QP with long-only bounds and getting "infeasible", then blaming the solver.
  - "Always check feasibility before solving; the right neutrality depends on the mandate."
- **Connections**: Part 3 (relative neutrality kills the factor bets in TE).

1.14 Index tracking: motivation and practice [S38–S40]

- **Definition**: a tracking portfolio reproduces a benchmark's return, ideally with fewer holdings, lower cost, and small, controlled deviations. "The central problem is benchmark-relative risk after holdings, costs, and time are accounted for" [S38].
- **Why optimise [S39]**:
  1. Deliver the benchmark: index funds and ETFs must reproduce the index return closely and cheaply.
  2. Holding every name is costly: thousands of constituents, tiny weights, turnover and liquidity all add up.
  3. So pick a smaller portfolio that minimises TE within the mandate's constraints. The small gaps in the return paths *are* the tracking error.
- **S&P 500 in practice [S40]**:
  - 500 names, cap-weighted. The top 10 are ≈ **a third** of the index; there is a long tail of ≈400 names, each **under 0.2%**.
  - Full replication is costly: hundreds of micro-positions, constant rebalancing, fractional-lot problems.
  - **Sample instead**: hold **~50–150 names** chosen to minimise TE.
  - Respect constraints: match sector weights, long-only, cap position sizes, limit turnover, fully invested.
- **Comparison [Beyond slides]**: full replication (TE ≈ 0, high cost) vs stratified sampling (match sector/factor cells) vs optimised sampling (min TE with $\|w\|_0 \le K$).
- ⚠ The top-10 share and tail sizes are illustrative and change over time. Quote them as "≈" in exams.

1.15 Active weights, active return, tracking error [S41–S42]

- **Definitions**:

$$a = w - b, \qquad e = a^\top \tilde{r} \ \text{(active return)}, \qquad \mathrm{TE} = \sqrt{a^\top \Sigma a} = \mathrm{sd}(e).$$

  - a holds your over- and under-weights. With $\mathbf{1}^\top w = \mathbf{1}^\top b = 1$, $\mathbf{1}^\top a = 0$: every overweight is funded by an underweight.
  - TE is "the portfolio volatility formula applied to a instead of to w". Small TE means you hug the index.
- **Units**: TE is annualised if Σ is annual. From monthly data, $\mathrm{TE}_{ann} = \sqrt{12}\,\mathrm{TE}_{mon}$ [Beyond slides].
- **Slide code [S42]**:
  - Setup: a 10-name benchmark `BENCH` (sector names such as Hlth and Enrgy; data not shown), holding 6 names with $w = (.22, .18, .20, .15, .15, .10)$ at positions 0, 2, 4, 6, 7, 9.

  - Output: active weights sum to $-1.4 \times 10^{-17} \approx 0$; biggest overweight Hlth **+5.0 pp**; biggest underweight Enrgy **−11.0 pp** (a dropped name, so $a_i = -b_i$).
  - **TE = 2.30% p.a.** Whether that is acceptable "is a mandate question, not a maths one".
- **Worked example [Beyond slides]**:
  - Inputs: Σ from 1.10 ($\begin{pmatrix} 0.0776 & 0.0384 \\ 0.0384 & 0.0356 \end{pmatrix}$), $w = (0.6, 0.4)$, $b = (0.5, 0.5)$, so $a = (0.1, -0.1)$.
  - $a^\top \Sigma a = 0.01(0.0776) - 2(0.01)(0.0384) + 0.01(0.0356) = 0.000776 - 0.000768 + 0.000356 = 0.000364$.
  - **TE = 1.91%**.
- **Interpretation**:
  - Dropped names contribute $a_i = -b_i$.
  - TE rises with the size of the bets and with *low* correlation between over- and under-weighted names. Highly correlated swaps are cheap in TE.
- **Traps**:
  - Computing TE as $\sqrt{w^\top \Sigma w} - \sqrt{b^\top \Sigma b}$ (difference of vols). That is wrong; use the vol of the difference.
  - Using $\sum |a_i|$ as a risk measure. That's "active share", not TE.
  - Ex-ante TE ($\sqrt{a^\top \Sigma a}$) vs ex-post TE (sample sd of realised e_t).
- **Implementation**:

```
a  = w - b
te = np.sqrt(a @ SIG @ a)          # annualised if SIG is annual
```

1.16 Tracking-error decomposition: factor bets + stock-specific [S43–S44]

- **Statement**:

$$\mathrm{TE}^2 = \underbrace{(B^\top a)^\top \Omega (B^\top a)}_{\text{factor bets}} + \underbrace{a^\top D a}_{\text{stock-specific}} .$$

  - **Factor bets**: your tilt away from the benchmark's factor exposures. Style or duration bets survive **even if sectors match**.
  - **Stock-specific**: risk from holding only a subset of names. More holdings shrink it, at higher trading cost.
- **Derivation ([S44], exam-reproducible)**:
  1. $\mathrm{TE}^2 = a^\top \Sigma a$ with $a = w - b$.
  2. $\Sigma = B \Omega B^\top + D$.
  3. Substitute: $a^\top (B \Omega B^\top + D) a$.
  4. Distribute: $a^\top B \Omega B^\top a + a^\top D a$.
  5. Group: $(B^\top a)^\top \Omega (B^\top a) + a^\top D a$.
  - The factor part depends only on the **active factor exposure** $B^\top a$. Set it to zero and it vanishes.
- **Link**: "Kill the hidden factor bets by matching exposures, $B^\top (w - b) = 0$." This is where factor neutrality meets index tracking (relative neutrality, 1.13). The remaining TE is $\sqrt{a^\top D a}$.
- **Worked example [Beyond slides]** (same data as 1.15):
  - $B^\top a = 1.2(0.1) + 0.8(-0.1) = 0.04$.
  - Factor part: $0.04^2 \times 0.04 = 0.000064$.
  - Specific part: $0.01(0.02) + 0.01(0.01) = 0.0003$.
  - Total: $\mathrm{TE}^2 = 0.000364$, so **TE = 1.91%** ✓ (matches the direct calculation). The factor share is 17.6%.
  - Matching beta ($B^\top a = 0$) would cut TE to $\sqrt{0.0003} = 1.73\%$.

- **Interpretation**:
  - Two books can have identical sector weights and still carry large factor TE (e.g. value/momentum tilts).
  - With many names, $a^\top Da$ shrinks roughly like $1/K$ (more holdings, smaller residual bets).
- **Traps**: forgetting that the factor term uses Ω (factor covariance), not σ_f or Σ.
- **Implementation**:

```
def te_split(w, b, B, Om, D):
    a = w - b; x = B.T @ a
    return x @ Om @ x, a @ D @ a        # (factor TE^2, specific TE^2)
```

1.17 Sparse replication: choosing K names out of N [S45]

- **Problem**:

$$\min_w (w-b)^\top \Sigma (w-b) + \tau \|w - w^-\|_1 \quad \text{s.t.} \quad \mathbf{1}^\top w = 1, \quad w \geq 0, \quad \|w\|_0 \leq K.$$

  - Objective: tracking error plus trading cost.
  - Constraints: fully invested, long-only, at most K names.
- **Intuition**:
  - Holding every name gives zero TE (take $w = b$) but is impractical. The TE-vs-K curve falls toward ~0 as $K \to N$, steep at first then flat.
  - The hard part is choosing **which** K to hold.
  - $\|w\|_0 \leq K$ makes the problem **nonconvex** (combinatorial: $\binom{N}{K}$ subsets).
- **Solution approaches [Beyond slides]**:
  1. **Greedy forward selection**: add the name that most reduces TE, re-optimise the weights, repeat until K.
  2. **Heuristic screen**: take the top-K by benchmark weight (or by sector/factor coverage), then re-optimise the weights on that subset.
  3. **MIQP**: binary z_i with $0 \leq w_i \leq z_i$ and $\sum z_i \leq K$ (exact, but slow for large N).
  4. **Stratified sampling**: pick names per sector/factor cell.

  - Given a subset, the weight problem is a **convex QP**.
- **Key trap [Beyond slides]**: an L1 penalty $\|w\|_1$ (lasso) **cannot** induce sparsity here, because with $w \geq 0$ and $\mathbf{1}^\top w = 1$, $\|w\|_1 = 1$ is constant. That's why $\|w\|_0$ appears explicitly.
- **Worked illustration [Beyond slides]** (simulated 20-stock, 1-factor universe; greedy with the long-only constraint relaxed within the subset): $K = 3$ gives TE 6.4%, $K = 8$ gives 2.4%, $K = 20$ gives 0. Diminishing returns in K.
- **Implementation (greedy) [Beyond slides, tested]**:

```
def track_subset(Sig, b, idx):                  # min (w-b)'Sig(w-b), 1'w=1, w=0 off idx
    Ss = Sig[np.ix_(idx, idx)]; c = (Sig @ b)[idx]; one = np.ones(len(idx))
    x, y = np.linalg.solve(Ss, c), np.linalg.solve(Ss, one)
    ws = x + (1 - one @ x) / (one @ y) * y       # Lagrange: budget multiplier
    w = np.zeros(len(b)); w[idx] = ws; a = w - b
    return w, np.sqrt(a @ Sig @ a)
def greedy(Sig, b, K):
    chosen = []
    for _ in range(K):
        j = min((j for j in range(len(b)) if j not in chosen),
                key=lambda j: track_subset(Sig, b, chosen + [j])[1])
        chosen.append(j)
    return chosen, track_subset(Sig, b, chosen)
```

Pitfall: this subset solver allows shorts inside the subset. For long-only, re-solve each subset with `scipy.optimize.minimize(method="SLSQP", bounds=[(0,1)]*n)` or a QP solver. Greedy is not globally optimal.

1.18 Synthesis, quiz and homework [S46–S48]

One page you can keep [S46]:

Model	The line that does it	Protects you from
Plain MVO	`np.linalg.solve(S, mu) / lam`	nothing; this *is* the problem
Ridge	`solve(lam*S + eta*I, mu)`	extreme, unstable weights
Robust	`w0 * (1 - kappa/t)`	a forecast you do not trust
Cost-aware	`gross - turnover * bps/1e4`	a book you cannot afford to trade
Factor-neutral	`w0 - (B@w0)/(B@Di@B) * (Di@B)`	a factor bet you did not mean to take
Index tracking	`sqrt(a @ S @ a), a = w - b`	drifting away from the mandate

"None of them is more than a few lines; the judgement is in knowing which mistake you are trying to avoid."

- ⚠ The "Robust" line omits the $\max(0,\cdot)$ floor shown in [S21].
- ⚠ The "Cost-aware" line only *evaluates* costs. The *optimisation* with $\tau\|w - w^-\|_1$ [S23] is what creates the no-trade region.

In-class quiz [S47]: a title slide only; no questions appear in the deck.

Group homework [S48]: build and stress-test a factor-neutral portfolio (one notebook per group).

1. **Estimate inputs**: $\hat{\mu}$, $\hat{\Sigma}$ from the return panel. Regress on the factors to get B, Ω, D, then assemble $\Sigma = B\Omega B^\top + D$.
2. **Naive MVO**: $w^* = \frac{1}{\lambda}\Sigma^{-1}\mu$. Nudge μ by a few bp and report how far w^* moves.
3. **Stabilise**: ridge $w^* = (\lambda\Sigma + \eta I)^{-1}\mu$. Tune η and report the drop in turnover and weight swings.
4. **Go factor-neutral**: $\max \alpha^\top w - \frac{\lambda}{2} w^\top D w$ s.t. $B^\top w = 0$. Check $B^\top w^* \approx 0$, and compare $\text{TE}^2 = a^\top \Sigma a$ across the three portfolios.

⚠ Step 4's TE comparison needs a benchmark b, which the slide doesn't specify. For zero-investment neutral books, $b = 0$ gives TE = total volatility; otherwise use the market/equal-weight portfolio.

---

2. Comparison tables

2.1 The three Part-1 fixes for estimation risk

	Plain MVO	Ridge (L2)	Robust (ellipsoid)	Cost-aware (L1 turnover)
Objective	$\mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$	$-\frac{\eta}{2}\|w\|_2^2$ added	$\min_{\tilde{\mu}\in U}$ of the MVO objective	$-\tau\|w - w^-\|_1$ added

	Plain MVO	Ridge (L2)	Robust (ellipsoid)	Cost-aware (L1 turnover)
Solution	$\frac{1}{\lambda}\Sigma^{-1}\mu$	$(\lambda\Sigma+\eta I)^{-1}\mu$	$w_0 \max(0, 1-\kappa/t)$ $(S=\Sigma)$	no closed form $(N>1)$; 1-D: clip to band
Knob	λ	η	κ $(=\varepsilon)$	τ
What it does to w	amplifies $\Delta\mu$ via Σ^{-1}	shrinks toward 0, **most in low-variance directions** (changes shape)	scales the whole tilt, **same direction**	anchors to w^-; creates a no-trade region
Exact zeros?	no	no	all zero if $\kappa \geq t$	yes, *trades* are exactly zero
Limit of knob $\to \infty$	—	$w \to \mu/\eta \to 0$ (1/N only with a budget and no signal)	$w=0$	never trade
Needs	Σ invertible	nothing extra	a choice of S, κ	w^- (drifted), cost model
Strength	simple, optimal if inputs are true	one line; stabilises; reduces turnover	principled "distrust" in SE units	realistic; cuts turnover directly
Weakness	error maximizer	η scale-dependent; tuned OOS	closed form only for $S \propto \Sigma$; otherwise SOCP	path-dependent; needs a solver
Use when	never raw on estimated inputs	many correlated assets, noisy μ	explicit forecast confidence	any live strategy / backtest

2.2 Constructions by mandate

	Absolute return MVO	Factor-neutral	Relative-neutral	Index tracking
Inputs	μ, Σ	α, D, B	α, D, B, b	Σ (or B, Ω, D), b, w^-
Constraint	none (cash absorbs)	$B^\top w = 0$	$B^\top(w-b)=0$	$\mathbf{1}^\top w = 1$, $w \geq 0$, $\|w\|_0 \leq K$
Solution	$\frac{1}{\lambda}\Sigma^{-1}\mu$	w_0 projected onto $B^\top w=0$ in the D-metric (§1.12)	same, with target $B^\top w = B^\top b$	nonconvex; greedy / MIQP
Earns	$\alpha^\top w + \beta_p^\top \lambda_f$	$\alpha^\top w$ only	benchmark return + $\alpha^\top a$	$\approx$ benchmark return
Risk	$w^\top \Sigma w$	$w^\top D w$	$a^\top D a$ active	TE^2
Long-only feasible?	yes	**no** if all $\beta_i > 0$	yes if $B^\top b$ is in the hull of the rows of B	yes (by design)

2.3 Risk decompositions

Quantity	Total	Factor part	Specific part
Portfolio variance [S5]	$w^\top \Sigma w$	$(B^\top w)^\top \Omega (B^\top w)$	$w^\top D w$
Tracking error² [S43]	$a^\top \Sigma a$	$(B^\top a)^\top \Omega (B^\top a)$	$a^\top D a$
Killed by	—	$B^\top w = 0$ / $B^\top a = 0$	more names (diversification)

2.4 Norms used this week

Norm	Formula	Role	Geometry / effect
$\|w\|_2^2$	$\sum w_i^2$	ridge size penalty	smooth; shrinks, spreads; no zeros
$\|w - w^-\|_1$	$\sum \|w_i - w_i^-\|$	trading cost / turnover	kink at 0, giving a no-trade region
$\|w\|_0$	# nonzero	cardinality (≤ K names)	nonconvex, combinatorial
$\|u\|_2 \le \varepsilon$	—	size of the mean uncertainty	ellipsoid via $S^{1/2}$

---

3. Master formula sheet

Factor form of a portfolio [S5]

- $\tilde{r} = \alpha + B\tilde{f} + \tilde{\varepsilon}$: the asset factor model (N×1; B is N×K).
- $\tilde{r}_p = \alpha_p + \beta_p^\top \tilde{f} + \tilde{\varepsilon}_p$, with $\alpha_p = w^\top \alpha$ and $\beta_p = B^\top w$.
- $E[\tilde{r}_p] = \alpha_p + \beta_p^\top \mu_f$, where $\mu_f = E[\tilde{f}]$.
- $\mathrm{Var}(\tilde{r}_p) = w^\top \Sigma w = \beta_p^\top \Omega \beta_p + w^\top D w$: factor risk + residual luck.
- $\Sigma = B\Omega B^\top + D$, which requires $\mathrm{Cov}(\tilde{f}, \tilde{\varepsilon}) = 0$ and a diagonal D [S30].
- $\mu = B\lambda_f + \alpha$ and $\alpha = \mu - B\lambda_f$; $D = \Sigma - B\Omega B^\top$ [S29].

Mandate objectives [S7]

- Absolute: $\max \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$.
- Neutral: $\max \alpha^\top w - \frac{\lambda}{2} w^\top D w$ s.t. $B^\top w = 0$.
- Tracking: $\min E[(w^\top \tilde{r} - \tilde{r}_b)^2]$ + costs, where $E[(a^\top \tilde{r})^2] = a^\top \Sigma a + (a^\top \mu)^2$ [Beyond].

Estimation risk [S8]

- $w^* = \frac{1}{\lambda}\Sigma^{-1}\mu$ and $\Delta w = \frac{1}{\lambda}\Sigma^{-1}\Delta\mu$.
- 2 assets, equal σ and correlation ρ [Beyond]: $\Delta w_1 = \frac{\Delta\mu_1}{\lambda\sigma^2(1-\rho^2)}$ and $\Delta w_2 = -\rho\,\Delta w_1$. With equal means m: $w_i = \frac{m}{\lambda\sigma^2(1+\rho)}$.

Ridge [S13-S17]

- $\|w\|_2 = \sqrt{\sum w_i^2}$.
- $\max \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w - \frac{\eta}{2}\|w\|^2$ gives $w^* = (\lambda\Sigma + \eta I)^{-1}\mu$.
- Eigen form [Beyond]: $w^* = \sum_j \frac{q_j^\top \mu}{\lambda d_j + \eta} q_j$.
- $\min \|w\|^2$ s.t. $\mathbf{1}^\top w = 1$ gives $w_i = 1/N$.

Robust [S20–S21]

- $U = \{\hat{\mu} + S^{1/2}u : \|u\|_2 \le \varepsilon\}$, equivalently $(\tilde{\mu} - \hat{\mu})^\top S^{-1}(\tilde{\mu} - \hat{\mu}) \le \kappa^2$.

- Worst case [Beyond]: $\min_U \tilde{\mu}^\top w = \hat{\mu}^\top w - \kappa\sqrt{w^\top S w}$.
- With $S = \Sigma$: $w^* = \frac{1}{\lambda}\Sigma^{-1}\hat{\mu} \cdot \max(0, 1 - \kappa/t)$, where $t = \sqrt{\hat{\mu}^\top \Sigma^{-1}\hat{\mu}}$.

Trading costs [S23–S24]

- $\max \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w - \tau\|w - w^-\|_1$.
- Turnover = $\sum |w^{new} - w^{old}|$ (one-way = ½ of it); cost = turnover × bps/10^4; net = gross − cost.
- Drift [Beyond]: $w_i^- = w_i(1 + r_i)/(1 + w^\top r)$.
- 1-asset band [Beyond]: $w^* = \text{clip}(w^-, \frac{\mu-\tau}{\lambda\sigma^2}, \frac{\mu+\tau}{\lambda\sigma^2})$.

Factor neutral [S27, S31–S34]

- Exposure $B^\top w = \sum_i w_i \beta_i$; neutral $\Leftrightarrow B^\top w = 0$; then $E[w^\top \tilde{r}] = w^\top \alpha$.
- $\mathcal{L} = \alpha^\top w - \frac{\lambda}{2} w^\top D w - \nu^\top B^\top w$ gives $\alpha - \lambda D w - B\nu = 0$.
- $\nu = (B^\top D^{-1} B)^{-1} B^\top D^{-1} \alpha$.
- $w^* = \frac{1}{\lambda}[D^{-1} - D^{-1}B(B^\top D^{-1}B)^{-1}B^\top D^{-1}]\alpha$.
- Projection form: $w_0 = D^{-1}\alpha/\lambda$ and $w^* = w_0 - D^{-1}B(B^\top D^{-1}B)^{-1}B^\top w_0$. For $K = 1$: $w^* = w_0 - \frac{B^\top w_0}{B^\top D^{-1} B} D^{-1} B$.
- Identity [Beyond]: $\alpha^\top w^* = \lambda w^{*\top} D w^*$.
- Relative neutral [S37]: $B^\top (w - b) = 0$.

Tracking [S41–S45]

- $a = w - b$ with $\mathbf{1}^\top a = 0$; $e = a^\top \tilde{r}$; TE = $\sqrt{a^\top \Sigma a}$.
- $\text{TE}^2 = (B^\top a)^\top \Omega (B^\top a) + a^\top D a$.
- Sparse: $\min (w - b)^\top \Sigma (w - b) + \tau\|w - w^-\|_1$ s.t. $\mathbf{1}^\top w = 1$, $w \geq 0$, $\|w\|_0 \leq K$.

---

4. Glossary

- **Active return (e)**: $a^\top \tilde{r}$, the portfolio return minus the benchmark return.
- **Active weights (a)**: $w - b$, the over- and under-weights vs the benchmark; they sum to 0.
- **Alpha (α)**: the expected return not explained by factor exposures (residual mean, selection skill); estimated as the regression intercept.
- **Benchmark (b)**: the reference portfolio (index) weights.
- **Beta-neutral / factor-neutral**: $B^\top w = 0$, zero net exposure to every factor.
- **bp / bps**: basis point = 0.01% = 0.0001.
- **Cardinality ($\|w\|_0$)**: the number of nonzero holdings.
- **Drifted weights (w^-)**: holdings after returns, before rebalancing.
- **Error maximizer**: MVO's tendency to put the largest bets where μ estimates are least reliable.
- **Estimation risk**: the risk that $\hat{\mu}, \hat{\Sigma}$ differ from the true values, and the portfolio damage that results.
- **Factor loadings (B)**: $N \times K$ sensitivities of asset returns to factors.
- **Factor covariance (Ω)**: $K \times K$ covariance of factor returns.
- **Factor premium (λ in [S29])**: the expected factor return; the reward for bearing factor risk.
- **Full replication**: holding every index constituent at index weight.
- **Gross exposure**: $\sum |w_i|$ (leverage).
- **Index tracking**: constructing a portfolio to reproduce a benchmark's return with low TE and cost.
- **L1 norm**: $\sum |x_i|$; used for turnover.
- **L2 norm**: $\sqrt{\sum x_i^2}$; its square is the ridge penalty.
- **Lagrange multiplier (ν)**: the shadow price of the neutrality constraints; equals the GLS projection of α on B.
- **Long-only**: $w \geq 0$.
- **Mandate**: the client's objective and constraints (absolute return, neutral, tracking).
- **MVO**: mean–variance optimisation.

- **NAV**: net asset value; weights and turnover are expressed as fractions of it.
- **No-trade region**: the set of current holdings where the L1 cost makes trading suboptimal.
- **Price of neutrality**: the alpha or utility given up by imposing $B^\top w = 0$ (0.029% in the [S35] example), plus the forgone factor premium.
- **Projection (risk-metric)**: the nearest feasible point to the unconstrained optimum, with distance $\|x\|_M^2 = x^\top M x$.
- **Reconstitution dates**: the dates an index changes membership or weights.
- **Relative neutrality**: $B^\top(w - b) = 0$, matching the benchmark's exposures.
- **Residual covariance (D)**: the diagonal covariance of idiosyncratic shocks.
- **Residual space**: working with α and D instead of μ and Σ once factor exposure is zero.
- **Ridge regularisation**: adding $\frac{\eta}{2}\|w\|^2$; equivalent to inverting $\lambda\Sigma + \eta I$.
- **Robust optimisation**: max–min over an uncertainty set of inputs.
- **Sampling (optimised / stratified)**: tracking with a subset of names.
- **Sparse replication**: tracking with at most K names.
- **Tangency portfolio**: the max-Sharpe risky portfolio, $\propto \Sigma^{-1}\mu$ (W2).
- **Tracking error (TE)**: $\sqrt{a^\top \Sigma a}$, the volatility of active return.
- **Turnover**: $\sum|\Delta w_i|$ (two-sided) or half that (one-way), as a fraction of NAV.
- **Two-fund separation**: every efficient portfolio is a combination of two funds (W2).
- **Uncertainty set (U)**: the set of plausible means; an ellipsoid here.
- t: $\sqrt{\hat\mu^\top \Sigma^{-1}\hat\mu}$, the forecast size (max Sharpe ratio).
- κ / ε: the radius of the uncertainty set (degree of distrust).
- η: the ridge strength.
- τ: the proportional trading-cost rate.

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5. Self-test

Questions

Q1 (conceptual, easy). Why is MVO called an "error maximizer"? In which *direction* of weight space does it amplify errors most, and what does such a position look like for two highly correlated stocks?

Q2 (easy). (a) Compute $\|w\|_2^2$ for $w = (0.5, 0.3, 0.2)$ and for $w = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$. (b) Prove that $\min \|w\|_2^2$ s.t. $\mathbf{1}^\top w = 1$ gives $w_i = 1/N$, and state the minimum value.

Q3 (numerical, easy). You rebalance from $w^{old} = (0.25, 0.25, 0.25, 0.25)$ to $w^{new} = (0.40, 0.30, 0.20, 0.10)$. The cost is 20 bp per unit traded and the month's gross return is 1.00%. Find $\sum|\Delta w|$, one-way turnover, cost and net return.

Q4 (numerical). Two assets have $\sigma = 25\%$, $\rho = 0.9$, $\lambda = 3$ and $\mu = (6\%, 6\%)$. (a) Find w^*. (b) Asset 1's forecast is revised to 7%. Find the new w^* and the change in each weight. (c) Without recomputing, how would the change in w_1 scale if $\rho = 0$?

Q5 (numerical). Two uncorrelated assets have $\sigma^2 = 0.04$ each, $\mu = (6\%, 4\%)$ and $\lambda = 4$. Compute MVO and ridge ($\eta = 0.04$) weights. By what factor does ridge shrink each weight? Why is the factor the same for both assets here but not in [S9]'s correlated example?

Q6 (derivation + numerical). (a) For $U = \{\hat\mu + S^{1/2}u : \|u\|_2 \le \kappa\}$, show that $\min_{\tilde\mu \in U} \tilde\mu^\top w = \hat\mu^\top w - \kappa\sqrt{w^\top S w}$. (b) With $S = \Sigma$, use the data of Q5 (no ridge) and $\kappa = 0.2$ to compute t, the shrink factor and the robust weights. (c) For what κ do you stop investing?

Q7 (numerical). One risky asset has $\mu = 6\%$, $\sigma = 20\%$ and $\lambda = 4$, and the L1 cost is $\tau = 50$ bp. Find the frictionless target and the no-trade band. What do you trade if current holdings are 0.30, 0.38 and 0.50?

Q8 (numerical, factor-neutral). One factor with $B = (1.0, 1.5, 0.5)^\top$, $D = \mathrm{diag}(0.04, 0.09, 0.01)$, $\alpha = (2\%, 1\%, 1.5\%)$ and $\lambda = 2$. Compute w_0, $B^\top w_0$, $B^\top D^{-1} B$ and the neutral w^*. Verify neutrality and compute the alpha before and after (the price of neutrality).

Q9 (conceptual). A long-only fund with stock betas $(0.8, 1.1, 1.3)$ is asked to be "market-neutral". (a) Show that this is infeasible. (b) Propose the correct constraint relative to an

equal-weight benchmark and check that it is feasible. (c) What premium does a truly neutral long–short book forgo?

Q10 (numerical, TE decomposition). One factor with $\Omega = 0.03$, $B = (1.5, 0.5)^\top$, $D = \mathrm{diag}(0.03, 0.02)$, $w = (0.3, 0.7)$ and $b = (0.5, 0.5)$. Compute a, the factor and specific TE² parts, and TE. What would TE become if you imposed $B^\top a = 0$ while keeping the same specific bets?

Q11 (derivation, hard). For $\max \alpha^\top w - \frac{\lambda}{2} w^\top D w$ s.t. $B^\top w = 0$: (a) derive ν and w^*; (b) show $\alpha^\top w^* = \lambda\, w^{*\top} D w^*$; (c) show that if $\alpha = Bc$ then $w^* = 0$, and interpret.

Q12 (hard, synthesis). (a) Show that $\max \mu^\top w - \frac{\lambda}{2} w^\top \Sigma w$ s.t. $B^\top w = 0$, with $\mu = B\lambda_f + \alpha$ and $\Sigma = B\Omega B^\top + D$, has the same solution as the residual-space problem. (b) Show that the unconstrained MVO objective equals $-\frac{\lambda}{2}(w - w_0)^\top \Sigma (w - w_0) +$ const, so constrained MVO is a projection. (c) Explain why a lasso penalty cannot replace $\|w\|_0 \le K$ in the long-only, fully-invested sparse-tracking problem.

Solutions

S1.

- $w^* = \frac{1}{\lambda}\Sigma^{-1}\mu$, so $\Delta w = \frac{1}{\lambda}\Sigma^{-1}\Delta\mu$.
- In eigen terms, a component of $\Delta\mu$ along eigenvector q_j is multiplied by $1/(\lambda d_j)$. The **smallest-eigenvalue (lowest-variance) directions** get the biggest multipliers.
- These are exactly the directions where μ is hardest to estimate relative to its effect, so the largest bets sit on the least reliable information.
- For two stocks with $\rho \to 1$, the low-variance direction is $(1, -1)$: a **large long-short pair**, long the one with the slightly higher estimated mean and short the other.

S2.

- (a) $0.25 + 0.09 + 0.04 = \mathbf{0.38}$; equal weights give $3 \times \frac{1}{9} = \mathbf{0.333}$.
- (b) $\mathcal{L} = w^\top w - \gamma(\mathbf{1}^\top w - 1)$, so $2w_i = \gamma$ for all i and all weights are equal. The budget gives $w_i = 1/N$ and $\|w\|^2 = N \cdot \frac{1}{N^2} = 1/N$. It is a minimum because the objective is strictly convex.

S3.

- $|\Delta| = (0.15, 0.05, 0.05, 0.15)$, so $\sum = \mathbf{0.40}$. One-way turnover = $\mathbf{0.20}$.
- Cost $= 0.40 \times 20/10^4 = \mathbf{0.08\%}$, so net $= 1.00 - 0.08 = \mathbf{0.92\%}$.
- If "20 bp per unit of one-way turnover" was meant instead, the cost would be 0.04%. State the convention.

S4.

- Setup: $\sigma^2 = 0.0625$ and $1 - \rho^2 = 0.19$.
- (a) $w_i = \frac{0.06}{3(0.0625)(1.9)} = \frac{0.06}{0.35625} = \mathbf{0.1684}$ each.
- (b) $\Delta w_1 = \frac{0.01}{3(0.0625)(0.19)} = \frac{0.01}{0.035625} = \mathbf{+0.2807}$ and $\Delta w_2 = -0.9(0.2807) = \mathbf{-0.2526}$. New $w^* = \mathbf{(0.449, -0.084)}$ (verified numerically).
- (c) With $\rho = 0$: $\Delta w_1 = \frac{0.01}{3(0.0625)} = 0.0533$, i.e. the change is $\mathbf{\div 5.26}$ (the factor $1/(1-\rho^2) = 5.26$ disappears).

S5.

- Σ is diagonal, so everything is per asset.
- MVO: $w_i = \mu_i/(\lambda\sigma^2) = \mu_i/0.16$, giving **(0.375, 0.25)**.
- Ridge: $w_i = \mu_i/(\lambda\sigma^2 + \eta) = \mu_i/0.20$, giving **(0.30, 0.20)**.
- Shrink factor $= 0.16/0.20 = \mathbf{0.8}$ for both.
- The factors are equal because $\lambda\Sigma$ has a single eigenvalue (0.16, isotropic), so every direction is shrunk equally.

- In [S9], the eigenvalues are 0.312 and 0.008 (times λ), so the spread direction is shrunk by 0.14 vs 0.86. Ridge changes the **shape** of w there.

S6.

- (a) $\tilde{\mu}^\top w = \hat{\mu}^\top w + u^\top S^{1/2} w$. By Cauchy–Schwarz, $u^\top S^{1/2} w \geq -\|u\|\,\|S^{1/2} w\| \geq -\kappa\sqrt{w^\top S w}$, with equality at $u = -\kappa S^{1/2} w / \|S^{1/2} w\|$. ■
- (b) $t^2 = \frac{0.06^2}{0.04} + \frac{0.04^2}{0.04} = 0.09 + 0.04 = 0.13$, so $t =$ **0.3606**. Shrink factor $1 - 0.2/0.3606 =$ **0.4453**. $w^* = (0.375, 0.25) \times 0.4453 =$ **(0.167, 0.111)**.
  - Expected return = $0.06(0.167) + 0.04(0.111) = 1.45\%$, vs $t^2/\lambda = 3.25\%$ nominal.
- (c) Stop investing when $\kappa \geq t = 0.3606$.

S7.

- $\lambda\sigma^2 = 0.16$, so the target is $0.06/0.16 =$ **0.375**.
- Band = $[\frac{0.055}{0.16}, \frac{0.065}{0.16}] =$ **[0.344, 0.406]**.
- $w^- = 0.30$: buy **4.4 pp to 0.344**, the edge, *not* the target.
- $w^- = 0.38$: **no trade**.
- $w^- = 0.50$: sell **9.4 pp to 0.406**.
- Reason: the marginal benefit of moving, $|\mu - \lambda\sigma^2 w|$, must exceed τ.

S8.

- $w_0 = D^{-1}\alpha/\lambda = (\frac{0.02}{0.04}, \frac{0.01}{0.09}, \frac{0.015}{0.01})/2 =$ **(0.250, 0.0556, 0.750)**.
- $B^\top w_0 = 0.25 + 0.0833 + 0.375 =$ **0.7083**.
- $D^{-1}B = (25, 16.67, 50)$ and $B^\top D^{-1} B = 25 + 25 + 25 =$ **75**.
- Scale = $0.7083/75 = 0.009444$.
- $w^* = w_0 - 0.009444(25, 16.67, 50) = (0.250 - 0.2361,\ 0.0556 - 0.1574,\ 0.750 - 0.4722) =$ **(0.0139, −0.1019, 0.2778)**.
- Check: $0.0139 - 0.1528 + 0.1389 = 0$ ✓.
- Alpha falls from $\alpha^\top w_0 = 0.005 + 0.000556 + 0.01125 =$ **1.681%** to $\alpha^\top w^* = 0.000278 - 0.001019 + 0.004167 =$ **0.343%**.
- The price of neutrality is 1.34%: most of this "alpha" was really aligned with beta.
- Check the identity: $\lambda w^{*\top} D w^* = 2(0.0000077 + 0.000934 + 0.000772) = 0.343\%$ ✓.

S9.

- (a) $\beta^\top w \geq \min_i \beta_i \cdot \mathbf{1}^\top w = 0.8 > 0$ for any long-only, fully invested w, so $\beta_p \in [0.8, 1.3]$ and 0 is unreachable.
- (b) Use $\beta^\top(w - b) = 0$. With $b = (\frac{1}{3}, \frac{1}{3}, \frac{1}{3})$, $\beta_b = 1.067 \in [0.8, 1.3]$, which is feasible (e.g. $w = b$, or any long-only w with $\beta^\top w = 1.067$).
- (c) The factor premium $\beta_p^\top \lambda_f$. A neutral book earns only $w^\top \alpha$ [S28].

S10.

- $a = (-0.2, 0.2)$.
- $B^\top a = -0.3 + 0.1 =$ **−0.2**.
- Factor part: $0.04 \times 0.03 =$ **0.0012**.
- Specific part: $0.04(0.03) + 0.04(0.02) =$ **0.0020**.
- $\text{TE}^2 = 0.0032$, so **TE = 5.66%** (verified against $\sqrt{a^\top \Sigma a}$ directly). The factor share is 37.5%.
- With $B^\top a = 0$, TE = $\sqrt{0.002} =$ **4.47%**.

S11.

- (a) $\alpha - \lambda D w - B\nu = 0$, so $w = \frac{1}{\lambda} D^{-1}(\alpha - B\nu)$. Then $B^\top w = 0$ gives $\nu = (B^\top D^{-1} B)^{-1} B^\top D^{-1} \alpha$, and $w^* = \frac{1}{\lambda}[D^{-1} - D^{-1}B(B^\top D^{-1} B)^{-1} B^\top D^{-1}]\alpha$.
- (b) Pre-multiply the FOC by $w^{*\top}$: $w^{*\top}\alpha - \lambda w^{*\top} D w^* - (B^\top w^*)^\top \nu = 0$. The last term is 0, so $\alpha^\top w^* = \lambda w^{*\top} D w^*$. ■
  - Consequence: the optimal utility is $\frac{1}{2}\alpha^\top w^*$.
- (c) If $\alpha = Bc$, then $\nu = (B^\top D^{-1} B)^{-1} B^\top D^{-1} B c = c$, so $\alpha - B\nu = 0$ and $w^* = 0$.

  - Interpretation: the "alpha" is a disguised factor tilt (proportional to loadings). Once factor bets are banned, there is nothing left to earn.

S12.

- (a) Under $B^\top w = 0$: $\mu^\top w = \lambda_f^\top B^\top w + \alpha^\top w = \alpha^\top w$, and $w^\top \Sigma w = (B^\top w)^\top \Omega (B^\top w) + w^\top D w = w^\top D w$.
  - The two objectives coincide on the entire feasible set, so they have the same maximiser.
  - (Their unconstrained optima differ, which is why the projections are in different metrics, yet they land on the same point.)
- (b) Let $w_0 = \frac{1}{\lambda}\Sigma^{-1}\mu$, so $\mu = \lambda \Sigma w_0$. Then

$$-\tfrac{\lambda}{2}(w-w_0)^\top \Sigma (w-w_0) = -\tfrac{\lambda}{2} w^\top \Sigma w + \lambda w^\top \Sigma w_0 - \tfrac{\lambda}{2} w_0^\top \Sigma w_0 = \mu^\top w - \tfrac{\lambda}{2} w^\top \Sigma w - \tfrac{\lambda}{2} w_0^\top \Sigma w_0.$$

  - So maximising the MVO objective over any feasible set = minimising the Σ-distance to w_0, i.e. a projection. ■
- (c) With $w \geq 0$ and $\mathbf{1}^\top w = 1$, $\|w\|_1 = \sum w_i = 1$ for *every* feasible w. The penalty is constant, so it cannot favour sparse solutions.
  - Hence the explicit (nonconvex) $\|w\|_0 \leq K$ constraint, solved by greedy selection, MIQP or heuristics.

Appendix A. Slide coverage checklist

Slide	Content	Section
S1	Title: Advanced Portfolio Construction	header
S2	Four skills (error maximizer, tame 3 ways, strip factor bet, track index)	§0
S3	Roadmap: 3 parts	§0
S4	Symbols ($w, \mu, \Sigma, B, \alpha, a$, TE)	Notation
S5	Portfolio in factor form, mean & variance split	1.1
S6	Week-2 handoff, new questions	1.2
S7	Three mandate objectives	1.2, 2.2
S8	$\Delta w = \frac{1}{\lambda}\Sigma^{-1}\Delta\mu$, error maximizer	1.3
S9	Worked example (ρ = 0.95, 64 pp)	1.3
S10	Code; ρ = 0.2 claim ⚠	1.3
S11	"How do we alleviate…"	1.5
S12	Size penalty picture	1.5
S13	L2 norm, (3,4) example	1.4
S14	Ridge formula	1.5
S15	Ridge derivation	1.5
S16	One big bet vs spread (1.00 vs 0.25)	1.5
S17	No signal gives 1/N	1.5
S18	Ridge code / η table	1.5

Slide	Content	Section
S19–S20	Robust: uncertainty set, max–min	1.6
S21	Robust closed form / κ table; ridge ≠ robust	1.6
S22–S23	Trading costs, L1, no-trade region, backtest charges	1.7
S24	Cost code: 13.22×, 2.64%, −1.44%; turnover ⚠	1.7
S25–S26	Factor-neutral intro, skill/factor/luck	1.8
S27	Beta-neutral $B^\top w = 0$ (+0.60, +0.24, −0.84)	1.9
S28	Cost of neutrality	1.9
S29	Residual space μ→α, Σ→D	1.10
S30	$\Sigma = B\Omega B^\top + D$ derivation, estimation	1.10
S31	Neutral problem	1.11
S32	Lagrangian closed form	1.11
S33	Constrained optimum = projection (risk metric)	1.12
S34	Neutralising = projection	1.12
S35–S36	Worked example & code (0.412, −0.618; 0.750% → 0.721%)	1.12
S37	Long-only infeasible, relative neutrality	1.13
S38–S40	Index tracking motivation, S&P 500 practice	1.14
S41–S42	Active weights, TE, code (2.30%)	1.15
S43–S44	TE decomposition & derivation	1.16
S45	Sparse replication, $\|w\|_0$ nonconvex	1.17
S46	One-page model table	1.18, 2.1
S47	In-class quiz (title only)	1.18
S48	Group homework pipeline	1.18

Appendix B. Flag summary (⚠)

1. **[S10]** At ρ = 0.2, the one-point nudge moves w_1 by **≈6.5 pp**, not "about 3".
2. **[S5] vs [S29]** μ means asset means and factor means; λ means risk aversion and the factor-premium vector. Disambiguate in answers.
3. **[S20] vs [S21]** The radius is written ε on one slide and κ on the other; they are the same thing. "κ in standard errors" holds only if S is the estimation-error covariance (Σ/T); the code uses $S = \Sigma$.
4. **[S24]** "One-way" turnover label: the code computes $\sum|\Delta w|$ (two-sided).
5. **[S7] vs [S41]** The tracking objective is the second moment $E[e^2] = \mathrm{TE}^2 + (E[e])^2$, while TE is the standard deviation.
6. **[S46]** The robust line omits $\max(0, \cdot)$; the cost-aware line evaluates costs rather than optimising with them.

7. **[S48]** The TE comparison needs a benchmark that the slide doesn’t specify.
8. K means the number of factors (Part 2) and the number of names held (Part 3).